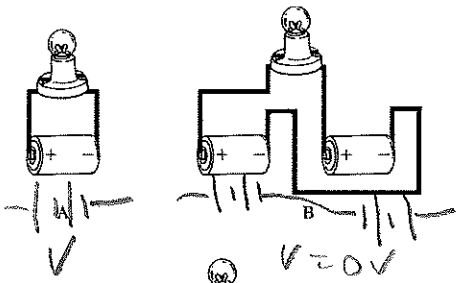
Practice Problems
Calculus Physics II

Name: _____ **Date:** _____

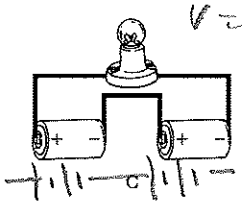
Chapter 44 More about Electric Circuits

1) Identical ideal batteries are connected in different arrangements to identical light bulb, as shown in the figure. For which arrangement will the bulb shine the *brightest*?



A) A
B) B
C) C
D) They all will have the same brightness.

2) For the above light bulbs, which light bulb will have no light?



A) A
B) B
C) C

3) For the above light bulbs, which light bulb will draw the most current?

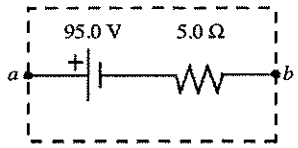
A) A
B) B
C) C
D) They all will draw the same amount of current.

4) A 12 V car battery (V_{emf} = 12 V) is wired in series the car's starter and draws 1.5 A of current, what is the internal resistance of the battery if when the car is being started the battery voltage is 8 V?

$V_T = emf - Ir$

$r = \frac{emf - V_T}{I} = \frac{12V - 8V}{1.5A} = 2.67\Omega$

5) The emf and the internal resistance of a battery are as shown in the figure. If a current of 8.3 A is drawn from the battery when a resistor *R* is connected across the terminals*ab* of the battery, what is the power dissipated by the resistor *R*?



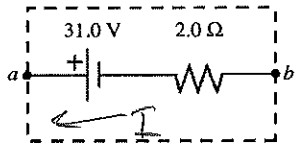
$P = I^2R = 344.4\ W$

$I = 8.3A$

$R = 5.0\Omega$

$P = IV = I^2R$

6) The emf and the internal resistance of a battery are as shown in the figure. When the terminal voltage V_{ab} is equal to 17.4 V, what is the current through the battery, including its direction?



$V_T = emf - Ir$

Solve for I

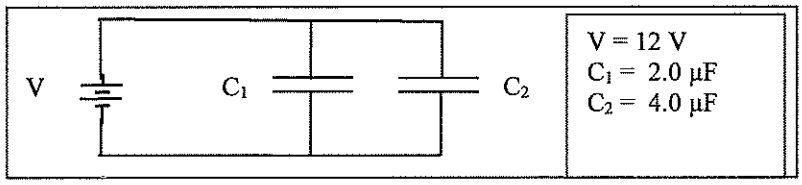
$I = \dfrac{emf - V_T}{r} = 6.8\,A$

$V_T = 17.4\,V$

$emf = 31\,V$

$r = 2\,\Omega$

7) Two capacitors are placed in parallel to a battery as shown, once the capacitors are charged.



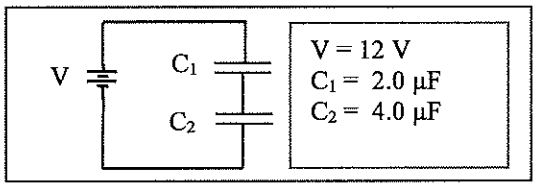
A) Determine the equivalent capacitance of the circuit?

$C_{eq} = C_1 + C_2 = 6\,\mu F$

B) Calculate the amount of charge on C_1?

$q_1 = C_1 V = (2\times10^{-6}\,F)(12\,V) = 2.4\times10^{-5}\,C$

8) Two capacitors are placed in series to a battery as shown, once the capacitors are charged



A) Determine the equivalent capacitance of the circuit?

$C_{eq} = \dfrac{1}{\left(\frac{1}{C_1} + \frac{1}{C_2}\right)} = 1.33\,\mu F$

B) Calculate the amount of charge on C_1?

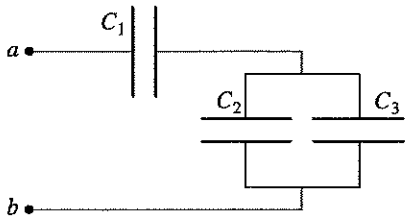
$q_1 = q_2 = q_{eq} = VC_{eq} = 1.6\times10^{-5}\,C$

C) V across C_1? $\quad V_1 = \dfrac{q_{eq}}{C_1} = \dfrac{1.6\times10^{-5}\,C}{2\times10^{-6}\,F} = 8\,V$

$q_1 = V_1 C_1$

9) Three capacitors are connected as shown to a 12 V voltage source between a and b.

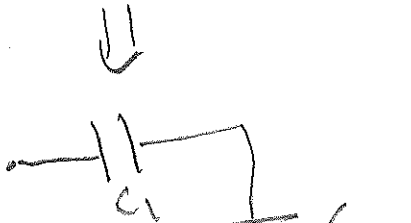
$C_1 = 4.0\ \mu F$
$C_2 = 3.0\ \mu F$
$C_3 = 2.0\ \mu F$



A) Determine the equivalent capacitance of the circuit.

$C_{23} = C_2 + C_3 = 5\,\mu F$

$C_{123} = \left(\frac{1}{C_1} + \frac{1}{C_{23}}\right)^{-1} = 2.222\,\mu F$



B) Determine the charge on each of the capacitors.

$q_{eq} = q_1 = q_{23} = C_{123}V = 2.67 \times 10^{-5}\,C$

$q = CV$

$V_{23} = \frac{q_{eq}}{C_{23}} = 5.33\,V$

$q_2 = C_2 V_{23} = 1.6 \times 10^{-5}\,C$

$q_3 = C_3 V_{23} = 1.07 \times 10^{-5}\,C$

10) A resistor, capacitor and dc power supply are connected in series with a switch. At the moment the switch is closed, the voltage across the capacitor is

no charge on capacitor yet

A) equal to the battery's terminal voltage.
B) less than the battery's terminal voltage, but greater than zero.
C) equal to the battery's terminal voltage.
(D) zero.

$V_{source} = V_R + V_C = I(t)R + \frac{q(t)}{C}$

11) A resistor, capacitor and dc power supply are connected in series with a switch. At the moment the switch is closed, the voltage across the resistor is

(A) equal to the battery's terminal voltage.
B) less than the battery's terminal voltage, but greater than zero.
C) equal to the battery's terminal voltage.
D) zero.

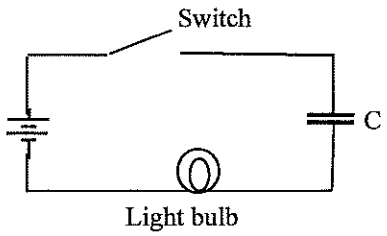
12) A resistor, capacitor and dc power supply are connected in series with a switch. After the capacitor is completely charged, the voltage across the resistor is

no current

A) equal to the battery's terminal voltage.
B) less than the battery's terminal voltage, but greater than zero.
C) equal to the battery's terminal voltage.
(D) zero.

13) A battery is connected in series with a switch, an un-charged capacitor and a light bulb, when the switch is closed what happens to the brightness of the light bulb over time?

A) It keeps getting brighter, then it stays the same brightness
B) It starts out bright then gets dimmer until it goes out
C) It starts out bright then gets dim until the capacitor is charged then it gets bright again
D) It is very dim until the capacitor is charged then it is bright
E) Once the switch is closed it turns on and never changes brightness



14) A 24 V power supply is wired in series with a 50 μC uncharged capacitor, a switch and a 400 kΩ resistor.

$V_0 = 24V$ | $\tau = RC = 20s$ | $q_0 = CV = 0.0012C$ |

A) When the switch is closed and the capacitor starts to charge, determine the amount of charge that is on the capacitor after t = 30 seconds.

$q(t) = q_0[1 - e^{-t/\tau}] = q_0[1 - 0.223] = 9.32 \times 10^{-4} C$

B) Determine the voltage across the capacitor after t = 30 s.

$V(t) = V_0[1 - e^{-t/\tau}] = 18.65V$

C) Determine the current flowing through the resistor at 30 s of charging.

$I_0 = \frac{V}{R} = 6.0 \times 10^{-5} A$

$I(t) = I_0 e^{-t/\tau} = 1.34 \times 10^{-5} A$

D) Calculate the voltage drop across the resistor at 30 s of charging.

$V_R = IR = 5.35V$ | or $V_R = V_0 - V_C = 5.35V$

E) How much time will it take to charge the capacitor to a voltage of 20 V?

$t = -RC \ln\left(\frac{V_0 - V}{V_0}\right) = 35.8 s$

15) An uncharged 5.0-μF capacitor and a resistor are connected in series to a 12-V battery and an open switch to form a simple RC circuit. The switch is closed at $t = 0$ s. The time constant of the circuit is 4.0 s.

A) Determine the value of the resistance R.

$\tau = RC$ | $R = \frac{\tau}{C} = 8 \times 10^5 \Omega$

B) Determine the maximum charge on the capacitor.

$q = CV_0 = 6 \times 10^{-5} C$

C) What is the charge on either plate after one time constant has elapsed? $t = \tau$

$q = q_0[1 - e^{-t/\tau}] = q_0[0.632] = 3.79 \times 10^{-5} C$

16) A 50 μC capacitor is charged to a potential of 24 V and is wired in series with, a switch and a 400 kΩ resistor.

$\tau = RC = 20s$ | $V_0 = 24V$ | $q_0 = CV_0 = 0.0012C$

A) When the switch is closed and the capacitor starts to discharge, determine the amount of charge that is on the capacitor after t = 30 seconds.

$q(t) = q_0 e^{-t/\tau} = q_0(0.223) = 2.68 \times 10^{-4} C$

B) Determine the voltage across the capacitor after t = 30 s.

$V(t) = V_0 e^{-t/\tau} = V_0(0.223) = 5.35V$

or

$V = \frac{q}{C} = 5.35V$

C) Calculate the voltage drop across the resistor at 30 s of discharging.

$V_s = V_C + V_R$

$V_R = V_s - V_C = 24V - 5.35V = 18.65V$

D) How much time will it take to discharge the capacitor to a voltage of 2 V?

$t = -RC\ln\left(\frac{V}{V_0}\right) = -20s\ln\left(\frac{2V}{24V}\right)$

$t = 49.7s$

KEY

1) C
2) B
3) C
4) 2.67 Ω
5) 344.4 W
6) 6.8 A
7) A) 6 μF B) 2.4×10^{-5} C
8) A) 1.33 μF B) 1.6×10^{-5} C C) 8 V
9) A) 2.22 μF B) $q_1 = 2.67 \times 10^{-5}$ C , $q_2 = 1.6 \times 10^{-5}$ C, $q_3 = 1.07 \times 10^{-6}$ C
10) D
11) A
12) D
13) B
14) A) 9.32×10^{-4} C B) 18.6 V C) 1.34×10^{-5} A D) 5.4 V E) 35.8 s
15) A) 8×10^{5} Ω B) 6×10^{-5} C C) 3.79×10^{-5} C
16) A) 2.7×10^{-4} C B) 5.4 V C) 18.6 V D) 50 s